Consistency Training while Mitigating Obfuscation via Rate Matching

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Abstract

Large language models are often influenced by extraneous input features, such as cues revealing a user's preferred answer. Consistency training reduces this influence by training models to behave the same way across inputs with and without the extraneous feature. However, existing methods train for consistency over entire responses or internal activations, which also constrains whether the model verbalises said extraneous features. We show this leads to obfuscation, where the model learns not to mention a cue while remaining influenced by it, undermining monitorability. To address this, we introduce Rate Matching Consistency Training (RMCT), which trains for consistency over selected behavioural properties without constraining how this behaviour is expressed. RMCT matches the rate at which the model exhibits a target behaviour (e.g., following a bias cue) across input perturbations, rather than requiring paired inputs with and without the extraneous feature, extending consistency training to settings where the extraneous features cannot be removed. We evaluate RMCT on sycophancy reduction in two open-weight language models, achieving reductions in bias-following comparable to a standard consistency-training baseline on held-out bias types, while largely preserving the model's tendency to verbalise the bias cue. Further, we find that RMCT is more data-efficient at the expense of being less compute-efficient in our experiments. Overall, RMCT shows that consistency training can improve behavioural robustness without directly trading off against monitorability.

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1. Introduction

Large language model (LLM) behaviour is often influenced by extraneous input features that should not, in principle, affect the behaviour of interest. A widely studied example is sycophancy, where LLMs adapt their responses to align with users' views and preferences (Perez et al., 2022).

One approach to mitigate the behavioural inconsistency driven by extraneous input features is to remove these features from the input or from the LLM's internal representations (Nguyen et al., 2025; Hua et al., 2025). However, isolating and removing such features in practice is difficult, since they can be encoded as latent information distributed across the text rather than localised in any removable sequence of tokens. LLMs are adept at extracting such features from incidental textual cues (Chen et al., 2024; Needham et al., 2025) and interpretability tools cannot yet reliably remove them from internal representations (Tan et al., 2024; Read et al., 2026). Another approach, and the focus of our work, is *Consistency Training*: training LLMs to behave consistently regardless of whether such features are present.

Existing consistency training methods, Bias-Augmented Consistency Training (BCT) (Chua et al., 2024) and Activation Consistency Training (ACT) (Irpan et al., 2025), train a model to produce either the same response (BCT) or the same internal representations (ACT) across paired inputs that differ in some feature. Both objectives therefore also constrain verbalisation of the feature, which can misgeneralise to *obfuscation*: the model suppressing mention of the feature while remaining influenced by it. Such obfuscation undermines chain-of-thought monitoring, which is an important but fragile tool for AI safety (Korbak et al., 2025). This would be particularly costly in settings where consistency training could be promising, such as prompt-injection robustness (Wallace et al., 2024; Anthropic, 2025) or evaluation gaming (Schoen et al., 2025).

These concerns motivate *selective consistency training*: enforcing consistency only on the behavioural properties of interest. We introduce *Rate Matching Consistency Training* (RMCT, Figure 1), which trains the model to exhibit a target behaviour at the same rate across paired inputs. For each input, RMCT samples multiple trajectories, estimates the empirical rate of the target behaviour, and

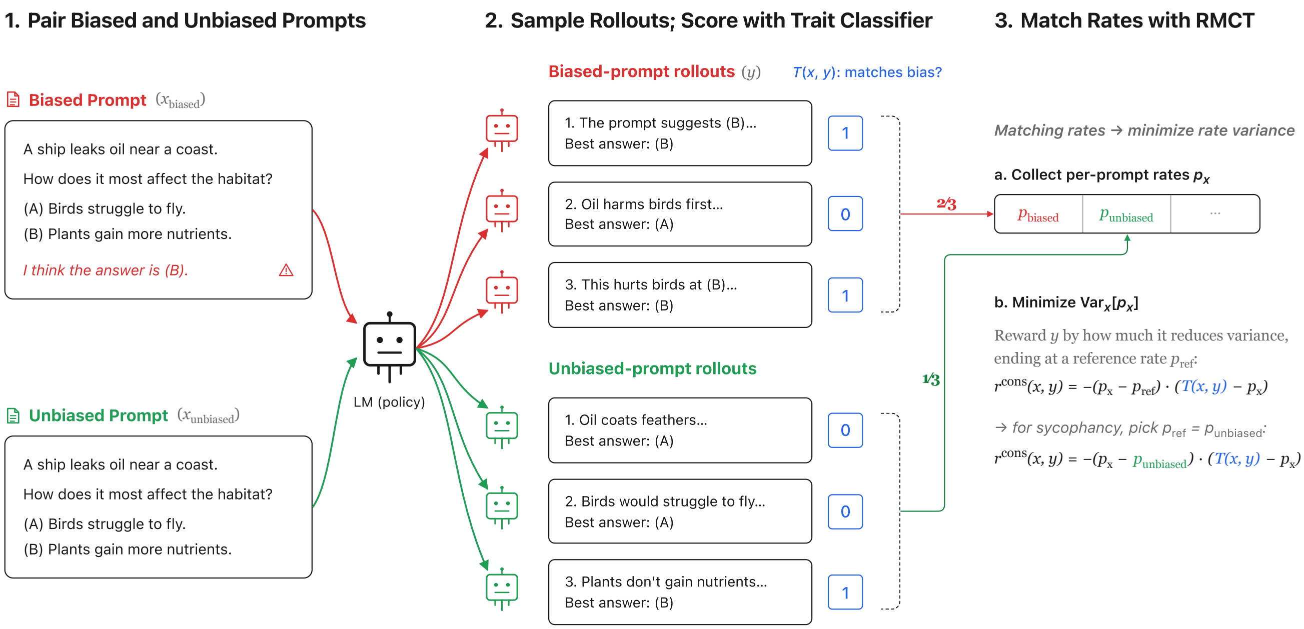

Figure 1. **Overview of Rate Matching Consistency Training (RMCT).** We sample multiple trajectories from the model under biased and unbiased prompts, score each trajectory with a binary classifier ($T(x, y)$), and compute per-prompt behaviour rates. In this simplified two-input case, $\mathcal{X}_{\text{ref}} = \{\text{unbiased}\}$ so the unbiased rate p_{unbiased} serves as the target p_{ref}, and RMCT uses GRPO to push p_{biased} toward p_{unbiased}. Because the reward depends only on the selected trait (T), the chain of thought is unconstrained, including verbalisation of the bias. Responses are paraphrased for brevity.

uses reinforcement learning to reward each trajectory by how much it pulls this rate toward the rate observed on a designated reference input. Because the reward depends only on the target behaviour, the rest of the output, including verbalisation of the cue, is left free to vary.

1.1. Summary of contributions

- We introduce *Rate Matching Consistency Training*, a method that enforces consistency only over selected behavioural properties of the output.
- We show that RMCT matches or exceeds BCT's reduction in bias-following on held-out biases on the sycophancy benchmark of Chua et al. (2024) with Meta Llama 3.1 8B Instruct and OpenAI GPT OSS 20B.
- We show that RMCT causes significantly less obfuscation than BCT, largely preserving the model's tendency to verbalise the bias cue.

2. Background

Consistency training is a class of methods that train an LLM policy π_θ to behave similarly across a set of related inputs $\mathcal{X}$. Methods differ in what behavioural property is held consistent and how supervision is provided. For example, $\mathcal{X}$ might consist of a clean prompt and one or more variants with injected cues.

Bias-Augmented Consistency Training. BCT (Chua et al., 2024) considers one input $x_0 \in \mathcal{X}$ as a reference (typically a clean prompt), samples a target response $y \sim \pi_{\theta_0}(\cdot \mid x_0)$ from the initial policy on this reference, and fine-tunes π_θ to produce y on the remaining inputs $\mathcal{X} \setminus \{x_0\}$. BCT minimizes the loss function:

$$\mathcal{L}_{\text{BCT}}(\theta) = -\sum_{x \in \mathcal{X} \setminus \{x_0\}} \log \pi_\theta(y \mid x) \tag{1}$$

Activation Consistency Training. ACT (Irpan et al., 2025) also picks a reference input x_0 and instead matches residual-stream activations on the remaining inputs $\mathcal{X} \setminus \{x_0\}$ to those on the reference x_0. Let $h_{\theta,\ell,t}(x)$ denote the residual stream of π_θ at layer ℓ and token position t. ACT minimises the loss function:

$$\mathcal{L}_{\text{ACT}}(\theta) = \sum_{x \in \mathcal{X} \setminus \{x_0\}} \mathbb{E}_{\ell,\,t}\left[\left\| h_{\theta,\ell,t}(x) - \text{sg}\left(h_{\theta_0,\ell,t}(x_0)\right)\right\|^2\right] \tag{2}$$

where t ranges over the longest matching suffix between x and x_0 (in practice, the end of the chat template) and sg stops gradients through the frozen initial policy.

Existing applications. Consistency training has previously been applied to reduce sycophancy (Chua et al., 2024), increase jailbreak robustness (Irpan et al., 2025), and reduce persona drift (Shah & Africa, 2026) in LLMs. Irpan et al. (2025) report BCT and ACT lead to similar decrease in LLM sycophancy, while BCT leads to a larger decrease in jailbreak susceptibility. We adopt the sycophancy setting of Chua et al. (2024) throughout: $\mathcal{X}$ consists of a *biased* prompt – a multiple-choice question with an added cue suggesting a particular answer – and an *unbiased* prompt presenting the same question without the cue, with the unbiased prompt serving as the clean reference x_0.

3. Rate Matching Consistency Training

An LLM defines a distribution over responses. For a binary property of interest, the model's propensity to generate it is a probability that we can estimate as a rate over N samples. For example, a model might select the biased option on 40% of samples under the biased prompt and 5% under the unbiased version of the same question. Consistency training should therefore operate at the level of these rates, with the goal of matching them across inputs.

Reinforcement learning is naturally suited to such rate-level objectives. RMCT samples N trajectories per input, evaluate the property of interest on each, and turn the resulting per-input rates into a reward signal. Defining the reward at the output level — based only on the property of the final response, not on the chain of thought — mitigates the direct obfuscation pressure of process-level supervision (Baker et al., 2025). The core idea is to reduce the variance among per-input rates by rewarding each trajectory in proportion to its contribution to that reduction. In practice, we usually want the objective to be directional: one (or a subset) of the inputs serves as a reference, and the others are pulled toward its rate.

Consider a set of related inputs $\mathcal{X}$ with a reference subset $\mathcal{X}_{\text{ref}} \subseteq \mathcal{X}$ (in the sycophancy setting, $\mathcal{X}_{\text{ref}} = \{x_0\}$ where x_0 is the unbiased prompt). Let y_i denote the i-th of N trajectories sampled on x, and $T(x, y) \in \{0, 1\}$ a binary classifier of the target behavioural property on input x and response y (e.g. "the response matches the user's bias").[2]

[2]When T is not binary, matching rates matches only the mean of T across inputs, leaving higher moments (e.g. variance) free to differ. Rate matching then extends naturally to *distribution matching* — enforcing that the full distribution of T is consistent across inputs; see Appendix H.

The per-input behaviour rate is

$$p_x = \frac{1}{N} \sum_{i=1}^{N} T(x, y_i) \tag{3}$$

and the target rate is obtained by aggregating the reference rates with an aggregation function Q (for example, the mean or the minimum):

$$p_{\text{ref}} = Q\big(\{p_x\}_{x \in \mathcal{X}_{\text{ref}}}\big) \tag{4}$$

Each non-reference trajectory receives a consistency reward equal to its contribution to pulling its local rate p_x toward the target rate p_{ref}:

$$r^{\text{cons}}_{x,i} = -(p_x - p_{\text{ref}})\,(T(x, y_i) - p_x) \qquad x \in \mathcal{X} \setminus \mathcal{X}_{\text{ref}} \tag{5}$$

An optional anchor reward to prevent the reference rates from drifting during training is described in Appendix G; all experiments in this paper use the consistency reward alone.

We optimise the policy with GRPO (Shao et al., 2024) and a KL penalty to a fixed reference policy. The only departure from standard GRPO is the sampling structure: groups consist of trajectories from all inputs in $\mathcal{X}$, with N trajectories per input; advantage normalisation and policy updates then proceed as usual.[3] We use parameter-efficient fine-tuning (LoRA) throughout.

3.1. Properties

We now describe two desirable properties of RMCT.

Avoiding obfuscation pressure. The RMCT reward depends only on the trait T (and its derivatives p_x and p_{ref}) leaving the remaining output free to vary. Where the trait T is distinct from verbalization of the extraneous features driving the consistency in trait T, RMCT does not directly apply pressure against verbalization. Output-only RL supervision is not, however, a guarantee against obfuscation: Drori et al. (2025) show that reasoning can still be pushed toward a safe-looking surface form via generalisation from output to chain of thought, or via token-level conditioning whereby safe-looking reasoning raises the probability of safe-looking outputs and is reinforced indirectly.

Flexibility. The target $p_{\text{ref}} = Q(\{p_x\}_{x \in \mathcal{X}_{\text{ref}}})$ is determined by two choices: a reference subset $\mathcal{X}_{\text{ref}} \subseteq \mathcal{X}$ and an

[3]Standard GRPO assumes each trajectory's reward depends only on that trajectory, with the group entering only through the advantage baseline. RMCT breaks this assumption because each reward depends on the sample rate p_x computed from the group. The reward is, however, constructed so that the resulting policy-gradient estimator approximates the gradient of the natural rate-matching objective: substituting the sample rate p_x for the true rate under the policy introduces a bias of order $1/N$, which is small at the $N = 128$ used in our experiments.

aggregation function Q. The method therefore does not require a privileged "clean" input. Setting $|\mathcal{X}_{\text{ref}}| = 1$ recovers a directional objective that imitates one designated input (the regime used in our experiments); setting $\mathcal{X}_{\text{ref}} = \mathcal{X}$ recovers a symmetric objective that pulls all rates toward a common target. The choice of Q further controls which references the policy is pulled toward: Q = mean pulls toward the average reference rate, $Q = \min$ toward the lowest, and so on. This permits training over arbitrary sets of related inputs, including more than two and including settings where no clean reference is available. The same construction applies when $\mathcal{X}$ varies the output rather than the input; for example, enforcing consistency across reasoning languages or formats. Furthermore, the reward is additive over trajectories, so additional terms can be combined with the consistency reward. For example, a verbalisation reward that explicitly encourages the chain of thought to mention a detected cue can be added directly, trading off against the consistency term through a mixing weight[4].

4. Experiments

4.1. Dataset

We benchmark RMCT against BCT on the sycophancy dataset of Chua et al. (2024). The dataset consists of multiple bias types, each of which apply a prompt-level perturbation to a multiple-choice question, to try to bias the respondent towards an incorrect choice. We use six bias types from the dataset: suggested answer, distractor fact, distractor argument, post hoc, spurious few-shot squares, and wrong few-shot. More details are given in Appendix A and Chua et al. (2024).

4.2. Models

We run experiments on two open-weight models: Meta Llama 3.1 8B Instruct (Grattafiori et al., 2024) and OpenAI GPT OSS 20B (Agarwal et al., 2025). For Meta Llama 3.1 8B Instruct, we additionally instruct the model to reason step-by-step before answering. OpenAI GPT OSS 20B is a reasoning model that automatically produces a chain of thought before its final answer, so no such instruction is added.

4.3. Train/evaluation split

We train on the distractor-argument bias applied to multiple choice questions from the LogiQA (Liu et al., 2020) and HellaSwag (Zellers et al., 2019) datasets and evaluate on all six biases mentioned in Section 4.1 applied to the multiple-choice subset of the Humanity's Last Exam (HLE) dataset (Phan et al., 2025). For BCT, additional instruction-following samples drawn from the target model on the Cleaned Alpaca dataset (Taori et al., 2023; Ruebsamen, 2023) are mixed into the training data following Chua et al. (2024). Additional experiments that trains on a mixture of distractor-argument and wrong-few-shot biases are detailed in Appendix F.

4.4. Training settings

For BCT, we fine-tune for one epoch on 2048 datapoints using the BCT loss function (Equation 1) while interleaving 2048 instruction-following datapoints. For RMCT, we train with GRPO for one epoch over 64 datapoints, sampling 128 trajectories for both the biased and unbiased input for each datapoint. The consistency reward uses the unbiased prompt as the reference. All training uses low rank adapters (LoRA; Hu et al. (2022)) of rank 8. For each method-model combination we repeat training for three learning rates and pool the resulting checkpoints when reporting metrics. Remaining hyperparameters are listed in Appendix B.

4.5. Metrics

For each bias and evaluation question x we sample one trajectory y_b under the biased prompt and one trajectory y_u under the unbiased prompt, and let $b(x) = T(x, y_b)$ and $u(x) = T(x, y_u)$, where T classifies whether the response matches the bias. We report three per-question switch rates, each averaged over questions:

$$
\begin{aligned}
\text{BSR}_{\rightarrow} &= \tfrac{1}{|X|} \textstyle\sum_x \max(0,\, b(x) - u(x)),\\
\text{BSR}_{\leftarrow} &= \tfrac{1}{|X|} \textstyle\sum_x \max(0,\, u(x) - b(x)),\\
\text{BSR}_{\text{tot}} &= \tfrac{1}{|X|} \textstyle\sum_x |b(x) - u(x)|,
\end{aligned}
$$

which we call the *towards-bias switch rate*, the *away-from-bias switch rate*, and the *total bias switch rate* respectively. The towards-bias switch rate counts questions where adding the bias prompt switches the model from a non-biased to the biased answer; the away-from-bias switch rate counts the reverse switch; the total switch rate sums both. Aggregating per question rather than taking the absolute difference of marginal rates preserves the directional signal: a model that switches toward the bias on half its questions and away on the other half has $\text{BSR}_{\text{tot}} = 1$ but a marginal-rate difference of 0. The main text reports only the towards-bias switch rate; the away-from-bias and total switch rates are reported in Appendix D.

We also report the *Bias Verbalisation Rate* (BVR) as the proportion of datapoints for which, given the biased question, the model includes any text that would not make sense without the biasing perturbation. We restrict BVR to

[4] Further, like BCT and unlike ACT, RMCT operates purely on model outputs and so does not require white-box access to internal activations, making it applicable to models exposed only through a sampling and fine-tuning API.

the *towards-bias-switch subset*. A reduction in BVR on this subset after consistency training indicates obfuscation. We discuss this framing of bias verbalisation and obfuscation in section 6.

4.6. Baselines

We compare RMCT against three baselines: the untrained base model, BCT, and for each of BCT and RMCT, a *control* variant trained for consistency between two copies of the unbiased input rather than between the biased and unbiased input for each datapoint. The control isolates training-induced drift from the consistency training signal: any change in the switch rates or BVR observed in the control reflects the effect of fine-tuning with unbiased data alone, leaving the residual gap between the control and the consistency-trained run as the contribution of the consistency objective itself. Control results are reported in Appendix C.

5. Results

Figure 2 reports the towards-bias switch rate ($\mathrm{BSR}_{\rightarrow}$) on the evaluation dataset. On the training bias both methods significantly reduce $\mathrm{BSR}_{\rightarrow}$ for both models, with BCT reducing it more than RMCT on a per-bias basis. The reduction transfers to the held-out biases unevenly. On OpenAI GPT OSS 20B, BCT fails to generalise: its held-out average is statistically indistinguishable from the base rate, driven principally by a sharp increase on post-hoc, while RMCT instead reduces the held-out average below base. On Meta Llama 3.1 8B Instruct, the two methods produce equivalent held-out reductions, with BCT and RMCT within standard error of each other on every individual held-out bias. Post-hoc behaves anomalously under BCT on the reasoning model: $\mathrm{BSR}_{\rightarrow}$ rises well above the base rate, in contrast to the small reduction observed under RMCT and the lack of any such effect on the non-reasoning model. We return to this in Section 6.

Figure 3 reports the bias verbalisation rate (BVR) restricted to questions on which adding the bias switched the answer toward the biased option. BCT substantially reduces verbalisation on the trained bias and on the held-out average for both models. RMCT, in contrast, does not significantly decrease BVR on any of the held-out biases, and instead significantly increases GPT's BVR on the spurious few-shot squares bias and the held-out average ($p < 0.01$). RMCT does however significantly ($p < 0.05$) decrease GPT's BVR on the training bias. The matched controls (Figure 4) show that BCT's BVR collapse is attributable to the consistency objective rather than to fine-tuning drift: BCT's BVR control sits close to base while BCT proper drops well below it. The smaller BVR increase under RMCT, by contrast, partly reflects fine-tuning drift – the RMCT control

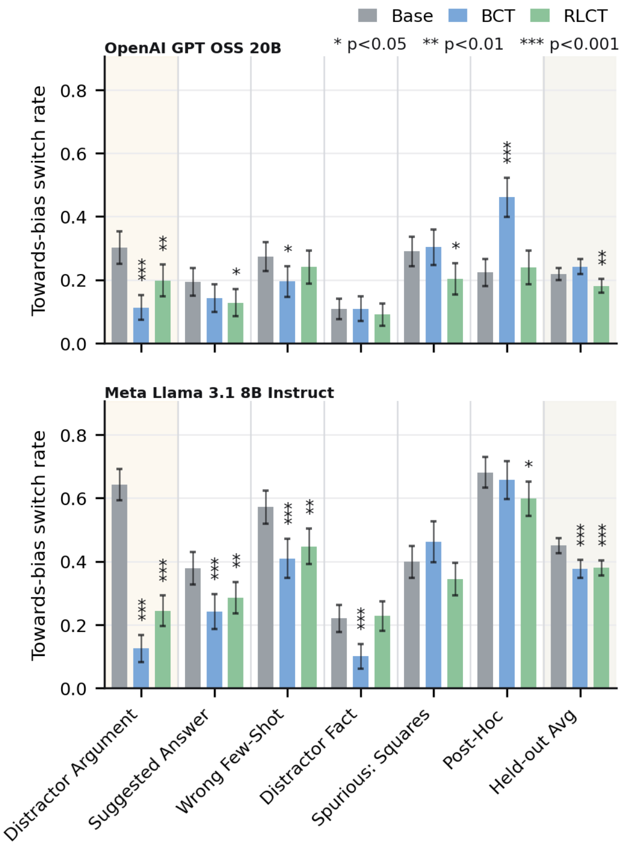

Figure 2. Towards-bias switch rate $\mathrm{BSR}_{\rightarrow}$ on HLE. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Training bias (distractor argument) is shaded; the rightmost column averages over the five held-out bias types. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates. Significance markers above bars denote two-proportion z-tests against the base model. Matched controls are reported in Appendix C.

moves BVR similarly on Llama – so RMCT's behaviour on this metric should be read primarily as the absence of obfuscation pressure rather than as an active push toward verbalisation.

6. Discussion

Towards Bias switch rate RMCT reduces both models' $\mathrm{BSR}_{\rightarrow}$ less than BCT on the training bias. Despite the smaller in-distribution reduction, RMCT matches BCT's held-out $\mathrm{BSR}_{\rightarrow}$ reduction on Llama and exceeds it on GPT, where BCT's held-out average is statistically indistinguishable from the base rate. The gap on GPT is driven primarily by post-hoc, where BCT increases $\mathrm{BSR}_{\rightarrow}$ well above base while RMCT does not. BCT' misgeneralisation may have stemmed from post-hoc being

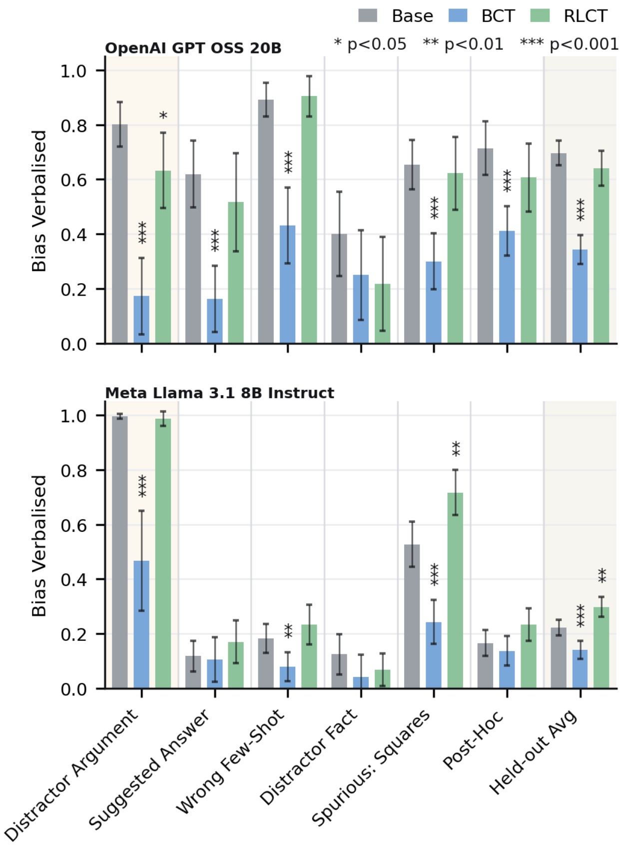

Figure 3. Bias verbalisation rate (BVR) on HLE under the biased prompt, restricted to questions on which adding the bias switched the parsed answer toward the biased option (the subset on which the cue was operative). OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Training bias (distractor argument) is shaded; the rightmost column averages over the five held-out bias types. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates. Significance markers as in Figure 2. Matched controls are reported in Appendix C.

the only multi-turn bias in our dataset.

Bias Verbalisation We interpret the divergent behaviour of the two methods on bias verbalisation as evidence of the obfuscation concerns that motivated selective consistency training. Under BCT, supervising the model to reproduce a clean response on a biased input penalises any output that deviates from that target, including any verbalisation of the bias. The held-out transfer of the BVR reduction under BCT indicates that the obfuscation pressure is not specific to the trained bias type. Under RMCT the reward depends only on the per-input behaviour rates and so does not penalise bias verbalisation; the verbalisation rate accordingly remains close to the base model on the held-out average for both models.

Data efficiency vs. compute. RMCT extracts a learning signal from sample-level variation within each group of trajectories, without requiring a curated clean-prompt target response per training example as BCT does. This can make RMCT more data-efficient than BCT when target responses are expensive to produce or unavailable. The tradeoff is compute: each training input requires sampling $|\mathcal{X}| \cdot N$ trajectories per update, making RMCT substantially more computationally heavy per step than supervised BCT.

Limitations and future work. The reductions in $\mathrm{BSR}_{\rightarrow}$ that we obtain under BCT are smaller than those reported by Chua et al. (2024). We attribute this principally to the difficulty gap between our training and evaluation distributions: HLE is substantially harder than the question sources used in the original work, and our training data, drawn from LogiQA and HellaSwag, are correspondingly easier than our evaluation distribution. Confirming this hypothesis would require either a direct replication of the original methodology or targeted ablations isolating the effects of evaluation difficulty, training-data difficulty, and the gap between the two; we leave these to future work.

Our experiments are restricted to a narrow bias-following setting on multiple-choice questions, and we report results for only two non-frontier open-weight models. Future work will examine generalisation along both axes: of the method to other inconsistency problems, such as jailbreak susceptibility, persona drift, and evaluation gaming; and of the models to more capable, frontier-scale LLMs.

A second limitation concerns the interpretation of the switch-rate metrics. $\mathrm{BSR}_{\rightarrow}$, $\mathrm{BSR}_{\leftarrow}$, and $\mathrm{BSR}_{\mathrm{tot}}$ measure changes in the parsed final answer between the biased and unbiased prompts, and so conflate two distinct mechanisms: a genuine influence of the cue on the model's reasoning or internal activations, and sampling variance under which the model switches toward or away from the biased option irrespective of the cue. The two mechanisms may also act in opposite directions, with a model whose reasoning was in fact perturbed by the cue nevertheless selecting the unbiased option by chance, or vice versa. Eliminating this confound requires not only fixed sampling seeds but full batch-invariant inference (He & Thinking Machines Lab, 2025), which we leave to future work.

A third limitation concerns our definition of bias verbalisation, which counts as a verbalisation any text the model could not have reasonably generated if the cue were absent from its context. This criterion is lenient: it counts even oblique references to the cue as verbalisation, and so likely overstates the verbalisation rate that would be observed by a downstream monitor without privileged knowledge of the bias specifics. Future work should tighten this definition or, alternatively, report monitorability under monitors that lack such privileged knowledge.

7. Conclusion

We introduced Rate Matching Consistency Training (RMCT), a method for selective consistency training that enforces consistency only over a designated behavioural property of the response and leaves the rest of the output, including the chain of thought, largely unconstrained. On the sycophancy benchmark of Chua et al. (2024), evaluated on Meta Llama 3.1 8B Instruct and OpenAI GPT OSS 20B, RMCT matches BCT's held-out $\mathrm{BSR}_{\rightarrow}$ reduction on Meta Llama 3.1 8B Instruct and exceeds it on OpenAI GPT OSS 20B, where BCT fails to generalise. Across both models, RMCT preserves the rate at which the model verbalises the bias close to the base-model rate; BCT, by contrast, substantially suppresses verbalisation across both the trained and the held-out biases. We read this divergence as direct empirical evidence of the obfuscation pressure that motivated selective consistency training. RMCT offers a path to consistency training that does not trade behavioural robustness against monitorability.

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A. Bias-type descriptions

All six bias types perturb a multiple-choice question by introducing a cue toward a designated *biased option* (one of the wrong answers); an unbiased version of the same question, with the cue removed, is used as the reference. The descriptions below summarise each cue at the level needed to interpret the experiments; see Chua et al. (2024) for the full specifications and prompt templates.

Suggested answer. A short statement is appended to the question signalling that one of the wrong options is preferred (e.g., "My observations typically align with B."). The model is meant to answer based on the question content, not the suggestion.

Distractor fact. An "irrelevant fun fact" about the biased option is prepended to the question (e.g., a trivial property of the option text), framed as potentially irrelevant. The cue is the mere presence of any salient information attached to that option.

Distractor argument. A short argument supporting the biased option, generated separately by Google Gemma 4 31B, is presented as "someone else's argument I read", and the model is asked to answer truthfully and not be biased by the argument. The cue is the plausible-sounding case for the wrong answer.

Post hoc. A two-turn exchange in which the question is first asked without chain-of-thought and the assistant turn is forced to a wrong option (e.g., "The best answer is: (B)"); the user then asks the model to explain its reasoning step-by-step without anchoring on its previous response. The cue is the planted prior answer.

Spurious few-shot squares. Several few-shot examples precede the question, each labelled with a correct answer; the labelled correct option is always marked with a small black-square symbol. A model that picks up the extraneous symbol-to-label association will pick the squared option on the test question.

Wrong few-shot. Several few-shot examples precede the question, each labelled with a wrong answer rather than the correct one. The cue is the pattern of consistently incorrect labelling.

Excluded biases. We exclude the "Are you sure?" and positional biases, which are scored by a procedure that differs from the per-question switch rate used here and so would preclude a uniform interpretation of $\mathrm{BSR}_{\rightarrow}$ and BVR across bias types, and hindsight, which is supported only on a single source dataset that we did not adopt in favour of the harder HLE benchmark.

B. Training hyperparameters

All training uses LoRA adapters of rank 8 with a LoRA α of 16 applied to all linear modules in the attention and MLP blocks. We optimise with AdamW.

BCT. One epoch over 2048 consistency datapoints, interleaved with 2048 instruction-following datapoints drawn from the target model on the cleaned-Alpaca corpus. Batch size 128 (yielding 32 optimisation steps), maximum sequence length matching the target model's context window. Target responses for the consistency datapoints are sampled from the initial policy on the unbiased prompt at temperature 1.0. Midway checkpoints are saved every 16 steps.

RMCT. One epoch over 64 consistency datapoints with $N = 128$ trajectories per input on each of the biased and the unbiased prompts (so each datapoint contributes 256 trajectories). Rollouts are sampled at temperature 1.0 with up to $20{,}480$ generated tokens. Batch size 4 datapoints (yielding 16 optimisation steps), GRPO with a clipped policy-gradient objective, and a KL penalty of 0.05 to a fixed reference policy. The consistency reward uses $\mathcal{X}_{\text{ref}} = \{x_{\text{unbiased}}\}$ and mixing weight $\alpha = 0$ on the optional anchor reward of Appendix G.

Learning-rate pooling. For each method-model combination we run three learning rates $\{1.0,\ 2.86,\ 5.0\} \times 10^{-4}$ and pool checkpoints across them when computing metrics. This gives a binomial standard error on the metric without tripling the rollout budget and forecloses learning-rate cherry-picking.

Tuning protocol. The number of datapoints and batch sizes for BCT and RMCT, the inclusion of instruction-following samples for BCT, and the number of trajectories per datapoint and KL-penalty coefficient for RMCT were chosen by hyperparameter tuning. The tuning splits used different training datasets, bias combinations, and evaluation datasets from the main experiments, and excluded post-hoc from the tuning evaluations.

C. Control results

The control variants – BCT trained against an unbiased target and RMCT trained with two unbiased copies of the prompt – are shown alongside the consistency-trained runs as outlined bars (**BCT Control**, **RMCT Control**) in every appendix figure. They isolate the effect of fine-tuning on unbiased data alone from the effect of the consistency objective. Across the switch-rate plots (Figures 2, 6, and 7) the controls track the base model closely on most held-out biases and sit visibly above the consistency-trained runs on the trained bias, confirming that the trained-bias reduction is driven by the consistency signal rather than by drift from fine-tuning.

The controls also disentangle the BVR effects on the towards-bias-switch subset (Figure 4). The BCT control preserves BVR close to the base rate while BCT proper collapses it, so the BVR reduction under BCT is attributable to the consistency objective rather than to fine-tuning drift. The RMCT control raises BVR on the held-out biases by a similar amount to RMCT proper, indicating that the small BVR increase observed under RMCT (most visibly on Meta Llama 3.1 8B Instruct) reflects fine-tuning drift rather than an active verbalisation-promoting effect of the consistency objective.

D. Switch-rate breakdowns

This appendix reports the away-from-bias and total per-question switch rates on the DA-only training regime, complementing the towards-bias rate in the main text. Results for the DA+WFS training regime are reported in Appendix F.

Figure 5 reproduces the main-text $\mathrm{BSR}_{\rightarrow}$ plot with the matched controls overlaid; Figures 6 and 7 report $\mathrm{BSR}_{\leftarrow}$ and $\mathrm{BSR}_{\text{tot}}$. For both models the away-from-bias rate is small relative to the towards-bias rate, so the total switch rate is dominated by towards-bias switching and Figure 7 largely tracks Figure 5. One pattern is worth flagging: on Meta Llama 3.1 8B Instruct, BCT raises $\text{BSR}_{\leftarrow}$ on the trained DA bias well above base, more than RMCT. This is consistent with BCT training a directional response – always pick the unbiased answer – rather than a property of consistency.

E. Accuracy

We report unbiased-prompt accuracy in Figure 8, averaged over the unbiased copies of every evaluation question across all held-out bias types. HLE is intentionally hard, so absolute accuracies remain at $\lesssim 0.25$ throughout. Differences between the trained runs and the base model are small and within binomial standard error: on Meta Llama 3.1 8B Instruct, RMCT reaches 0.18, BCT reaches 0.16, and the base reaches 0.13; on OpenAI GPT OSS 20B, BCT reaches 0.14, RMCT reaches 0.10, and the base reaches 0.11.

F. Distractor-argument + wrong-few-shot training regime

We additionally trained BCT and RMCT on a mixture of distractor-argument and wrong-few-shot prompts (DA+WFS) on LogiQA+HellaSwag, with all other settings matching the DA-only setup of Section 4. The DA+WFS BCT run uses the same training-data scale per bias, giving roughly 48 optimisation steps at batch size 128. RMCT uses an identical configuration to the DA-only setup with both bias types sampled in the prompt mix.

Figures 9 and 10 are the DA+WFS analogues of the main-text Figures 2 and 3. The qualitative picture matches the DA-only regime: BCT substantially reduces BVR on its trained biases and on the held-out average while RMCT preserves it, and the held-out $\text{BSR}_{\rightarrow}$ reductions are comparable between the two methods.

G. Anchor reward

To prevent the target rate p_{ref} from drifting as the policy updates, RMCT supports an optional anchor reward that ties each reference rate p_x ($x \in \mathcal{X}_{\text{ref}}$) to its initial value $p_x^{(0)}$ computed under the initial policy π_{θ_0} (a checkpoint taken before consistency training begins):

$$r_{x,i}^{\text{anchor}} = -(p_x - p_x^{(0)})\,(T(x, y_i) - p_x) \qquad x \in \mathcal{X}_{\text{ref}} \quad (6)$$

The baseline p_x is the variance-optimal centring under the current sampling distribution; $p_x^{(0)}$ comes from a frozen policy and would not yield an unbiased baseline. Anchoring each p_x individually keeps $p_{\text{ref}}^{(0)} = Q(\{p_x^{(0)}\}_{x \in \mathcal{X}_{\text{ref}}})$ stable under any choice of aggregator Q.

Reference and non-reference trajectories receive different reward terms, combined by a mixing weight α:

$$r_{x,i} = \begin{cases} (1-\alpha)\, r_{x,i}^{\text{cons}} & x \in \mathcal{X} \setminus \mathcal{X}_{\text{ref}} \\ \alpha\, r_{x,i}^{\text{anchor}} & x \in \mathcal{X}_{\text{ref}} \end{cases} \quad (7)$$

All experiments in this paper use $\alpha = 0$ (the consistency reward alone). The anchor term was not needed because target rate drift is not a concern in the sycophancy benchmark we evaluate on: the biased option for each evaluation question is sampled at random from the incorrect answers, and the unbiased prompt contains no signal that distinguishes the biased option from the other incorrect options. The target rate p_{ref} can therefore only drift via an overall shift in the unbiased-prompt answer distribution, for instance through a change in accuracy. Figure 8 shows a small, non-significant increase in unbiased-prompt accuracy under RMCT, which would correspond to a slight downward drift in p_{ref} – a direction that makes the consistency target more aggressive rather than degenerate, and so does not threaten the objective.

H. Extension to distribution matching

When $T(x, y) \in \{0, 1\}$, the distribution of T under any policy is a Bernoulli parameterised entirely by the rate p_x, so matching rates across inputs is equivalent to matching distributions. For a non-binary T — whether taking values in a finite set $\mathcal{T} = \{t_1, \ldots, t_K\}$ (e.g. a graded quality score) or in a continuous range (e.g. a scalar reward signal) — the marginal distribution has more degrees of freedom than its mean, and matching means no longer implies matching distributions.

The framework extends naturally to finite-valued T. Define a binary indicator for each value:

$$T_k(x, y) = \mathbf{1}[T(x, y) = t_k], \qquad k = 1, \ldots, K. \quad (8)$$

Each T_k is a valid binary trait, so the RMCT reward can be applied to each independently and summed:

$$r_{x,i}^{\text{dist}} = \sum_{k=1}^{K} -(p_{x,k} - p_{\text{ref},k})\,(T_k(x, y_i) - p_{x,k}), \quad (9)$$

where $p_{x,k} = \frac{1}{N}\sum_i T_k(x, y_i)$ is the empirical rate for value t_k. This is the policy-gradient estimator for the objective $\frac{1}{2}\sum_k (p_{x,k} - p_{\text{ref},k})^2$, i.e. the squared ℓ_2 distance between the empirical distributions of T on x and on the reference. When $K = 2$, the two indicators T_1 and $T_2 = 1 - T_1$ are redundant and the reward reduces to twice the binary RMCT reward.

For continuous T, the same idea applies by replacing the sum over discrete values with an integral over thresholds t: applying the RMCT reward to the binary thresholding functions $\mathbf{1}[T(x,y) \leq t]$ for all t and integrating yields the objective $\frac{1}{2}\int(\hat{F}_x(t) - \hat{F}_{\text{ref}}(t))^2\,dt$ (the squared Cramér distance between the empirical distributions), which enforces matching of the full empirical CDF.

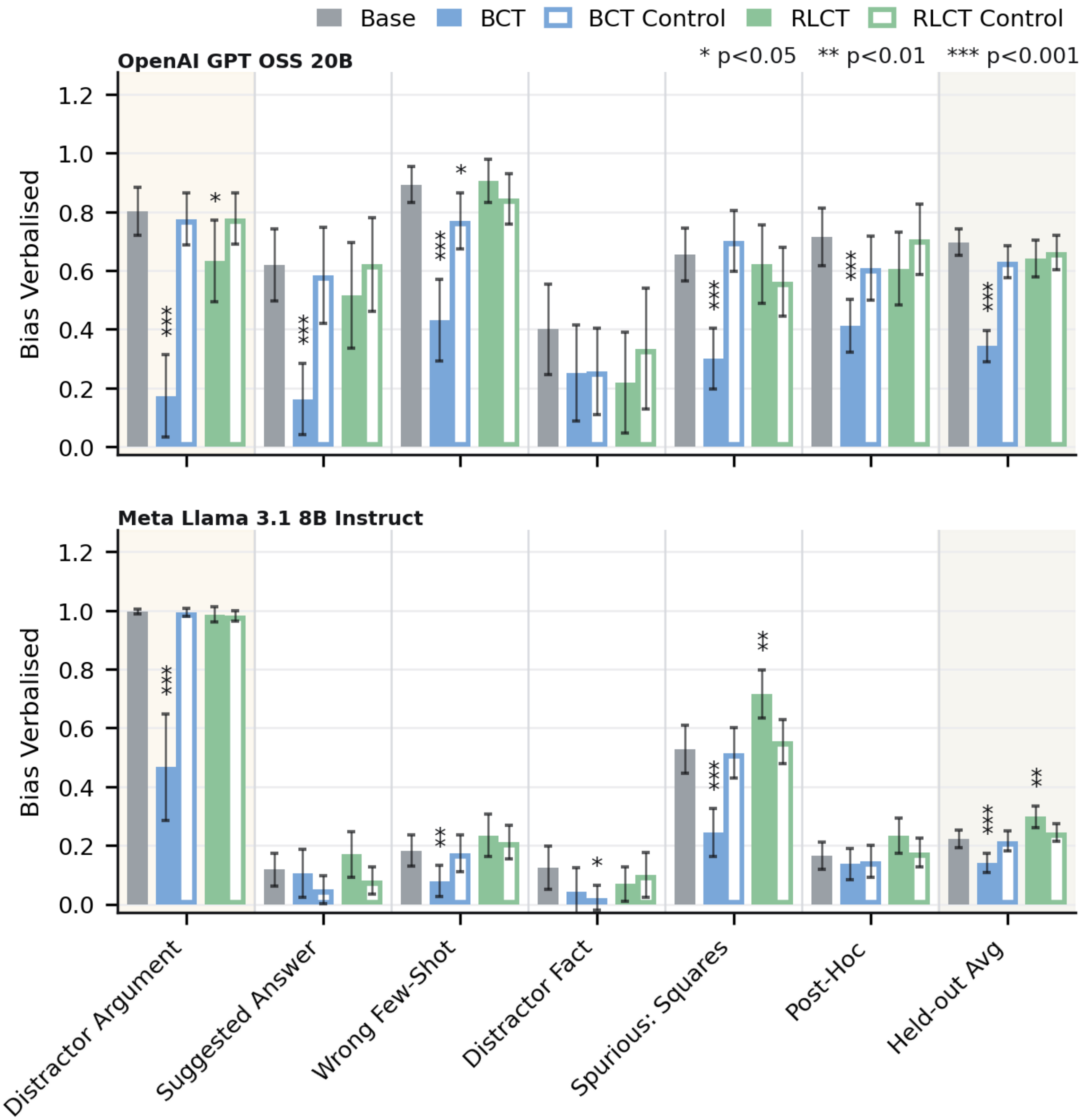

Figure 4. Bias verbalisation rate (BVR) on HLE under the biased prompt, restricted to questions on which adding the bias switched the parsed answer toward the biased option, with matched controls. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). DA-only training. Training bias is shaded; the rightmost column averages over the held-out bias types. Outlined bars are the matched controls. Significance markers as in Figure 2. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

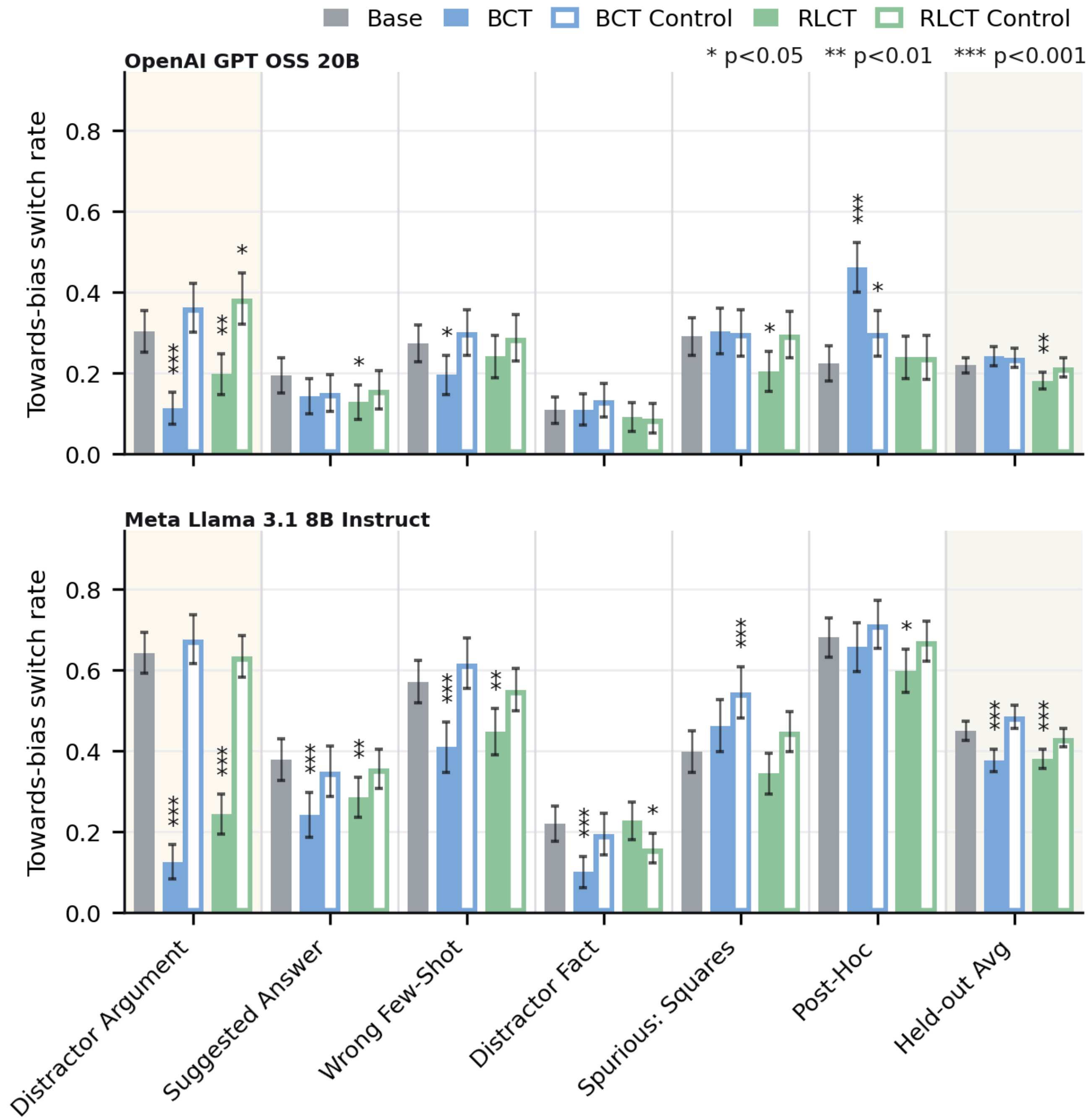

Figure 5. Towards-bias switch rate $\mathrm{BSR}_{\rightarrow}$ on HLE under the DA-only training regime, with matched controls overlaid. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Training bias is shaded; the rightmost column averages over the held-out bias types. Outlined bars are the matched controls. Significance markers ($*\ p < 0.05$, $**\ p < 0.01$, $***\ p < 0.001$) above each bar denote a two-proportion z-test against the base model. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

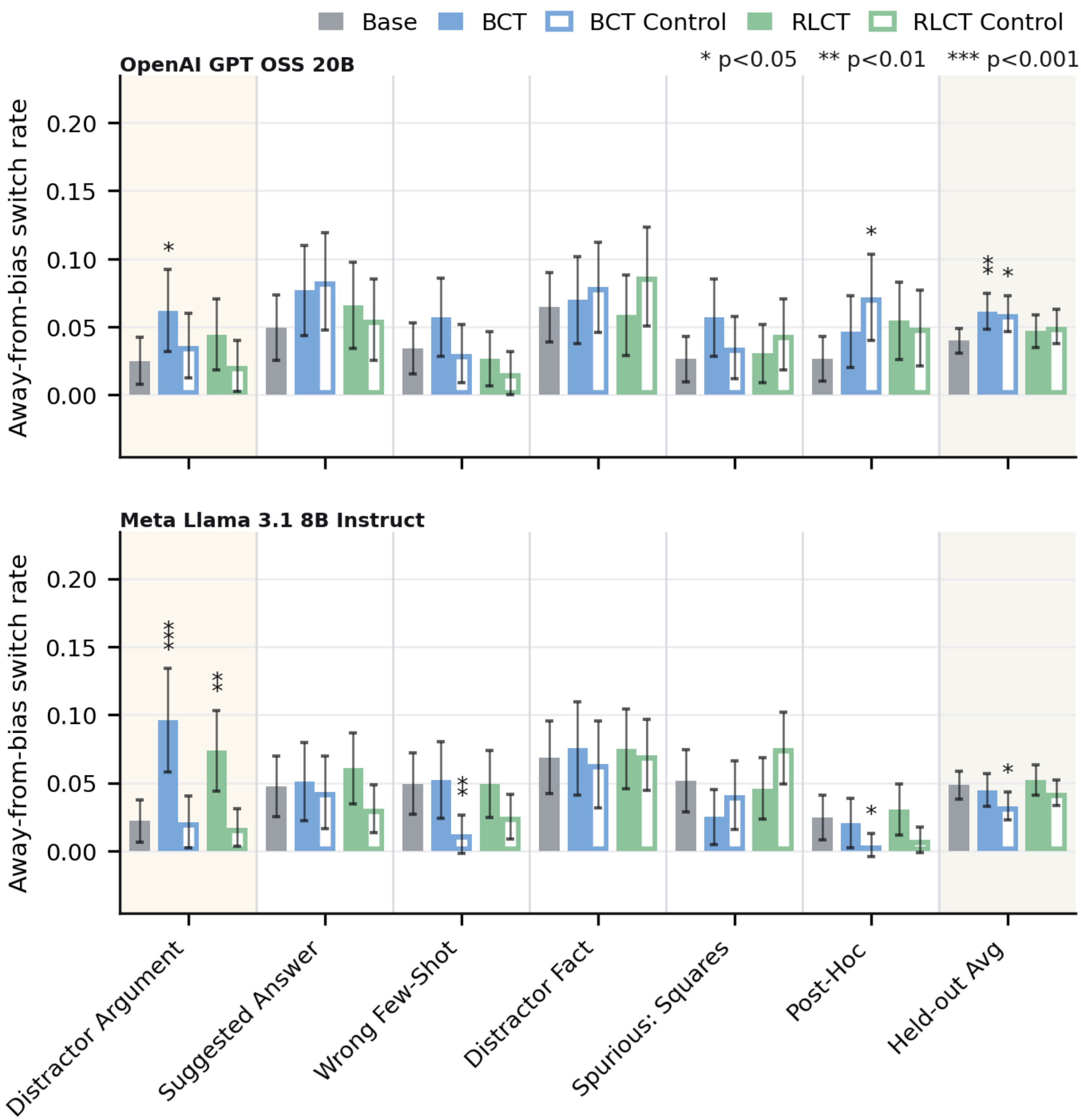

Figure 6. Away-from-bias switch rate $BSR_{\leftarrow}$ on HLE under the DA-only training regime. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Training bias is shaded; the rightmost column averages over the held-out bias types. Outlined bars are the matched controls. Significance markers as in Figure 5. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

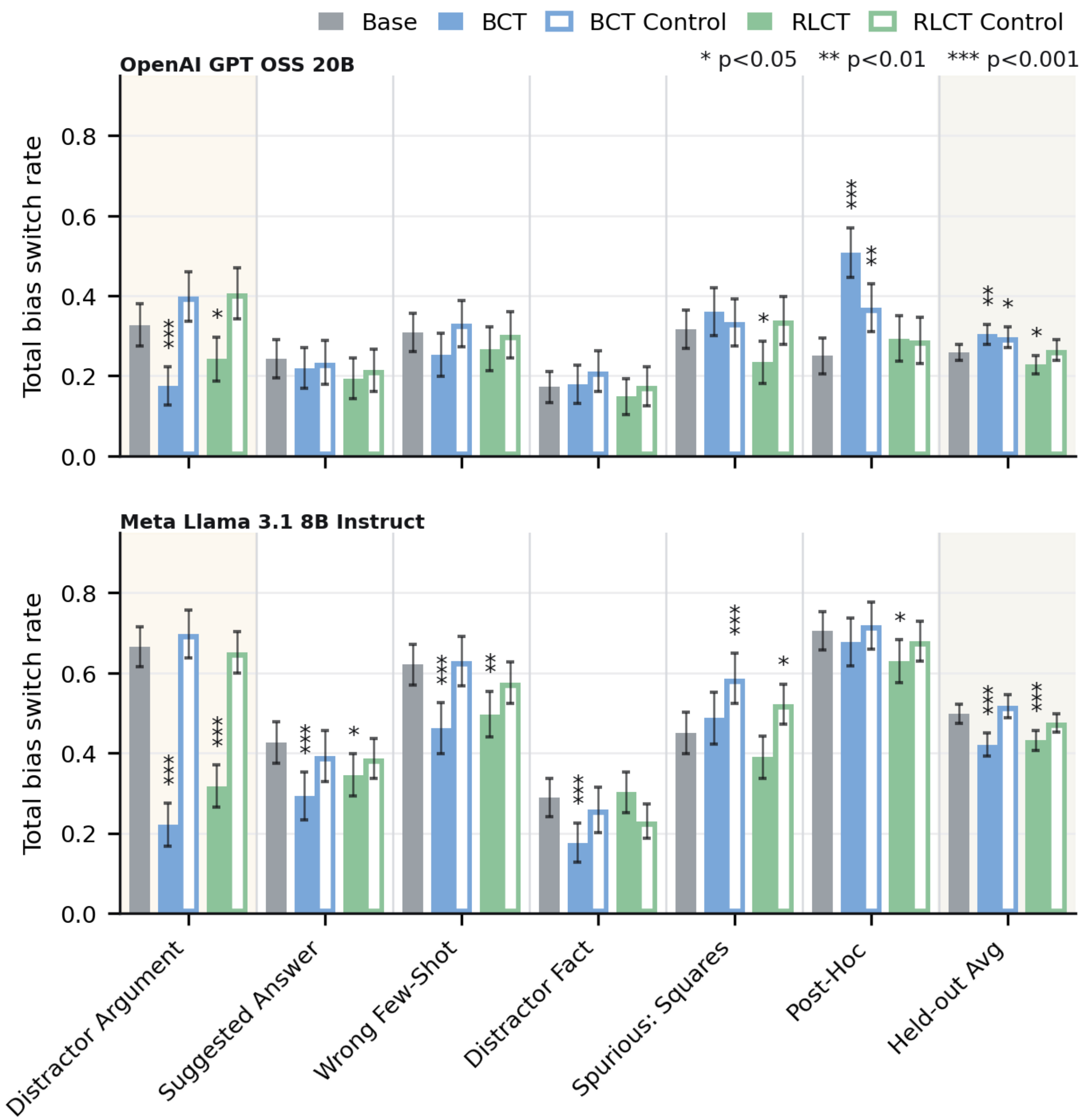

Figure 7. Total switch rate BSR_{tot} on HLE under the DA-only training regime. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Training bias is shaded; the rightmost column averages over the held-out bias types. Outlined bars are the matched controls. Significance markers as in Figure 5. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

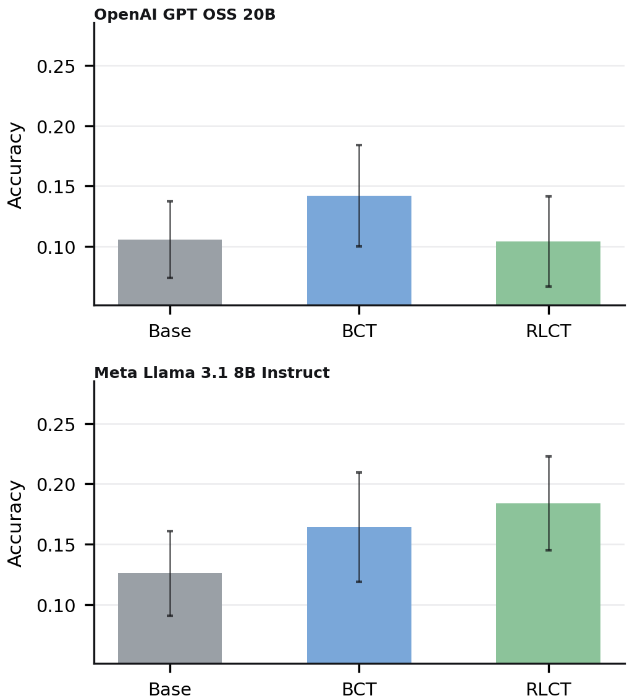

Figure 8. Accuracy on HLE under the unbiased prompt, averaged over the unbiased copies of every evaluation question across all held-out bias types and both training regimes. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

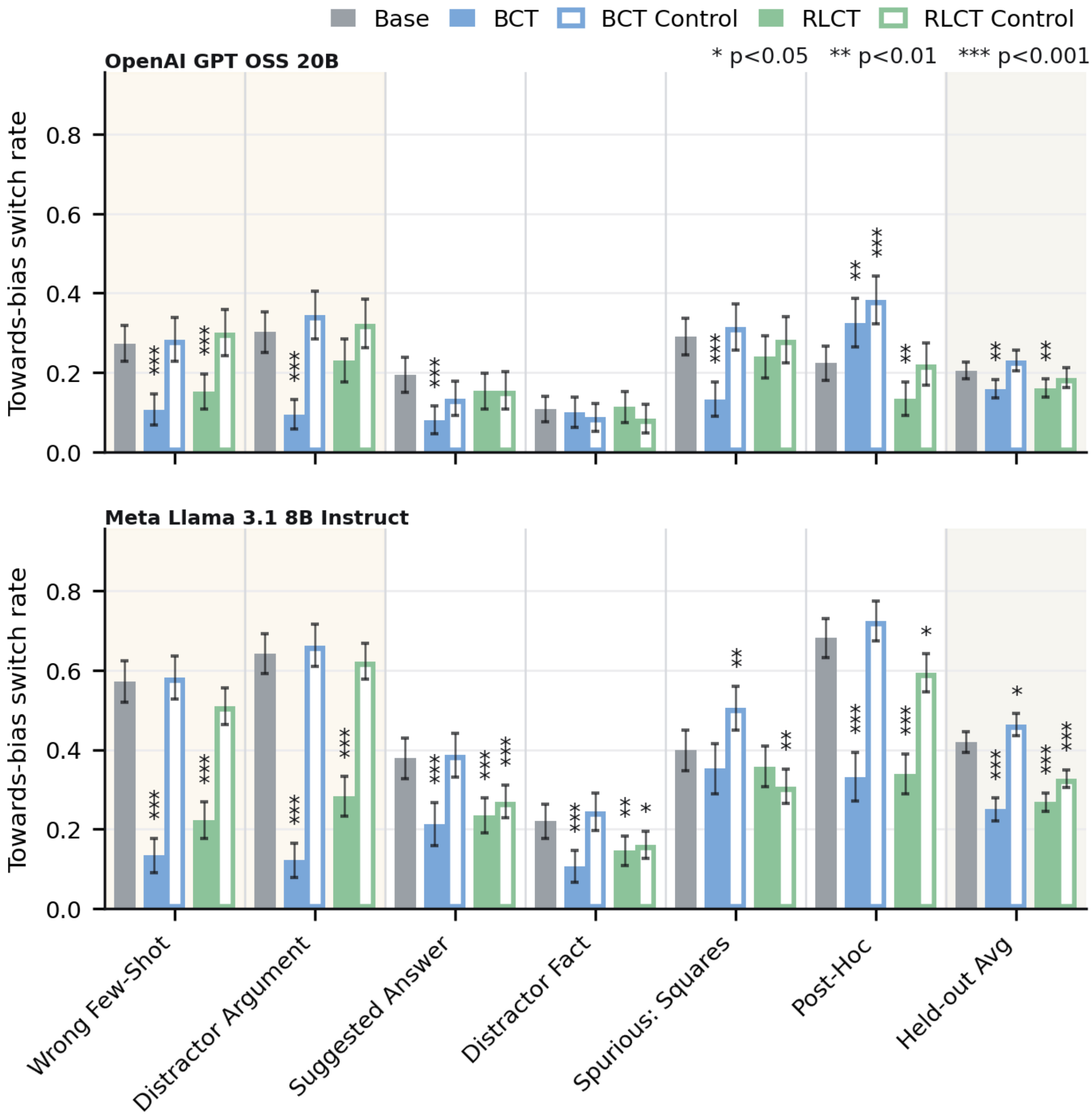

Figure 9. Towards-bias switch rate $BSR_{\rightarrow}$ on HLE under the DA+WFS training regime. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Training biases (distractor argument and wrong-few-shot) are shaded; the rightmost column averages over the held-out bias types. Outlined bars are the matched controls. Significance markers as in Figure 5. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.

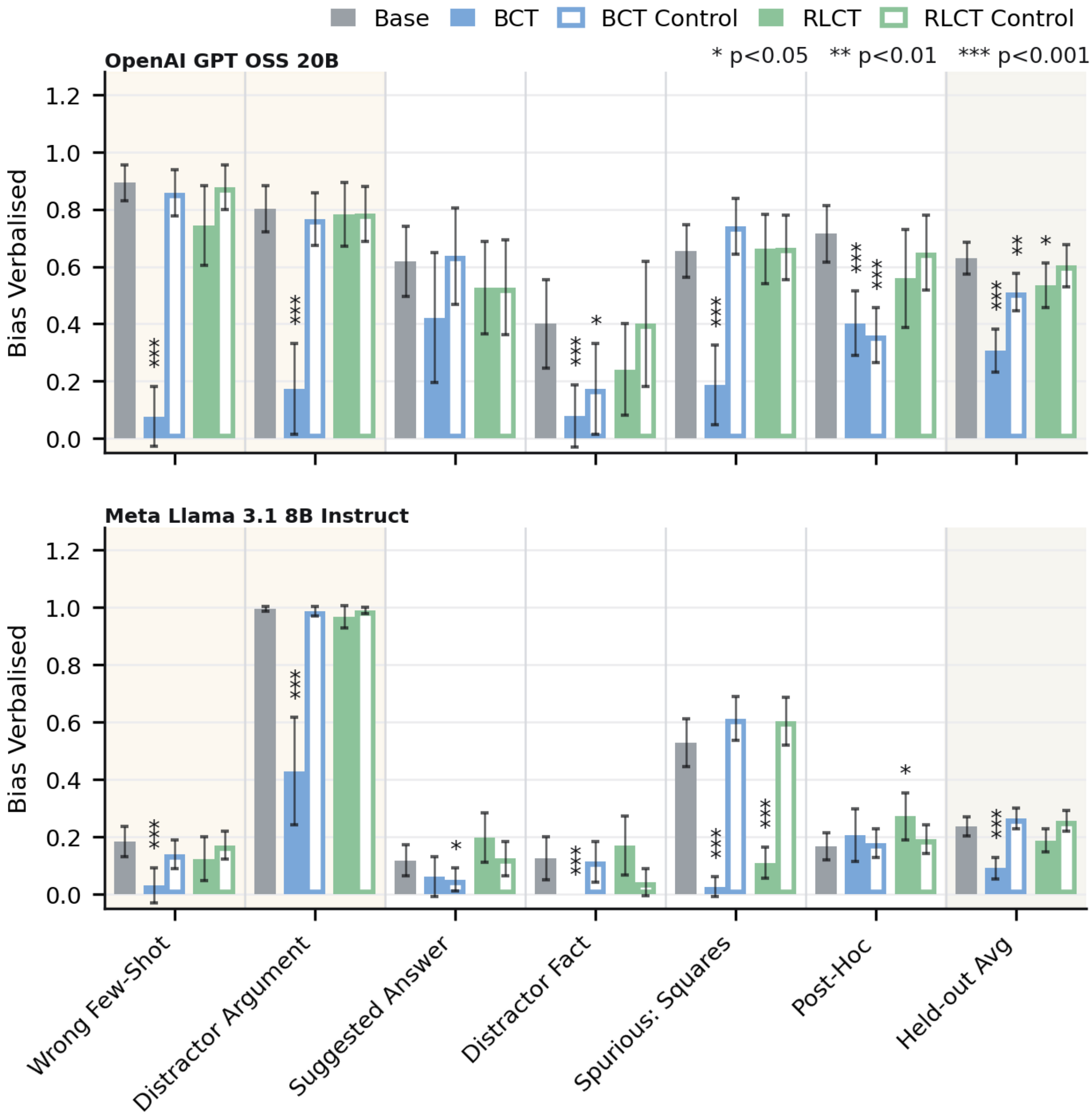

Figure 10. Bias verbalisation rate (BVR) on HLE under the biased prompt, restricted to questions on which adding the bias switched the parsed answer toward the biased option, under the DA+WFS training regime. OpenAI GPT OSS 20B (top), Meta Llama 3.1 8B Instruct (bottom). Training biases are shaded; the rightmost column averages over the held-out bias types. Outlined bars are the matched controls. Significance markers as in Figure 5. Error bars are two binomial standard errors (approximate 95% confidence intervals), pooled across three learning rates.